

MEEG 44403 Final Project

Brain Tumor Image Segmentation using Deep Learning

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December 10th 2025

Abstract

In this project, a fully integrated machine learning pipeline was developed to automate brain tumor analysis using multimodal MRI data. The model performs two primary tasks, 3D tumor segmentation and then prediction of overall patient survival. A 3D Attention UNet architecture was trained to segment key tumor subregions, including the enhancing tumor, edema, and necrotic core using T1, T1-CE, T2, and FLAIR MRI sequences from the BraTS 2020 dataset [1]. Following segmentation, radiomic features describing the tumor volume were extracted and then combined with the clinical metadata to train a fully connected neural network to for a survival regression model. The segmentation model demonstrated stable convergence and produced clinically meaning full Dice scores across the tumor subregions, while the survival network achieved strong alignment when it came to predictions and true survival outcomes. These results show that the pipeline was able to successful learn the different structures and related imaging biomarkers to patient prognosis. This work also establishes clear engineering relevance by demonstrating how computational models can support diagnostic workflows, treatment planning, and prognostic evaluation. The work presented here provided a solid foundation for future engineering development towards real time clinical decision support systems.

Introduction

In modern medicine, automated analysis of medical imaging plays a very critical role in improving diagnostic accuracy, treatment planning, and clinical decision making. One of the most challenging problems in this field is the ability to achieve completely reliable readings of MRI scans. Modern treatment planning for glioma patients relies very heavily on the accurate portrayal of tumor subregions, the

extraction biomarkers, and the ability to make predictions such as patient survival. Developing computational methods for identifying these tumors can provide significantly useful. Creating reliable predicts remains a significant engineering challenge due to the complexity of gliomas, variability in MRI acquisition protocols, and the limited amount of annotated clinical data.

To address these complexities, recent research has focused on integrating some advanced machine learning approaches, specifically with deep neural networks. In the segmentation task, 3D U-Nets, encoder-decoder architecture, and hybrid convolutional models have become standard due to their strong performance on medical data. These models aim to produce predictions that are able to differentiate enhancing tumor, necrosis, edema, and any other clinically relevant structures. The effectiveness oof the model depends on the preprocessing pipelines, proper handling multimodal inputs, and techniques for improving model generalization. T1 provides structural anatomy, T1-CE highlights enhancing tissue, T2 reflects fluid, and FLAIR emphasizes edema each corresponding to a different physical tissue property critical to engineering interpretation.

Beyond segmentation, an equally important task in tumor analysis is the overall prediction of patient overall survival. Unlike segmentation, survival prediction requires features from both imaging data and patient metadata. Machine learning models such as Random Forest, and fully developed networks attempt to map these tumor features to clinically meaningful outcome categories. The challenge, however, lies in designing models that are able to avoid overfitting while capturing the relationship between the tumor and the patient outcome. This establishes a strong engineering motivation accurate survival prediction directly supports

surgical planning, treatment selection, and resource allocation.

The engineering problem addressed in this work is the development of a machine learning based system is capable of performing accurate segmentation of brain tumor subregions and predicting clinically relevant outcomes such as overall patient survival after surgery.

The system under study consists of volumetric MRI data obtained through T1, T1-CE, T2, and T2-FLAIR sequences provided by Kaggle through the BraTS2020 challenge. These imaging modalities provided the physical signals from which tumor structures must be extracted. All imaging data had annotations provided by experienced neuro-radiologists with a segmentation mask manually erected on the MRI images. Annotations comprised of the GD-enhancing tumor (ET – label 4), the peritumoral edema (ED – label 2), and the necrotic and non-enhancing tumor core (NCR/NET – label 1).

The inputs of this system are multimodal MRI imaging data and clinical patient meta data. The Success metrics were deterred by the dice similarity coefficient for segmentation accuracy and the mean squared error for classification accuracy for survival prediction. These metrics ensure that performance can be objectively evaluated in an engineering context.

Methods

This work consisted of a multistage computational pipeline designed to preprocess multimodal MRI volumes, perform voxel wise tumor segmentation using a 3D convolutional neural network, extract clinically meaningful radiomic features from the segmentation mask, and predict patient survival using a regression model. All experiments were implemented with Python and trained using NVIDIA L4 GPU allowing full reproducibility. The following subsections describe the mathematical formula

of each stage. Each patient scans in the BraTS MRI dataset contain four modalities. Those being FLAIR, aT1, T1-CE, and T2. Below denote the 3D intensity fields used for MRI image preprocessing and Spatial Normalization. Where each I_m corresponds to a different MRI sequence.

MRI Preprocessing and Spatial Normalization

$$I_m(x, y, z), m \in \{1, 2, 3, 4\} \quad (1)$$

As MRI volumes vary in shape across patients, a standardized spatial size is required for 3D convolutional neural networks. Each scan is therefore resized to a fixed tensor shaped. As shown below, using a symmetric crop or pad operation.

$$T = (H^*, W^*, D^*) = (128, 128, 128) \quad (2)$$

For an input dimension H , the crop indices are defined as

$$s_H = \max\left(0, \frac{H-H^*}{2}\right) \quad (3)$$

$$e_H = \min(s_H + H^*, H) \quad (4)$$

and the placement indices on the output tensor are

$$c_H = \max\left(0, \frac{H^*-H}{2}\right) \quad (5)$$

$$c'_H = c_H + (e_H - s_H) \quad (6)$$

This operation is applied identically along all spatial axes, producing a normalized volume of.

$$\tilde{I}_m(x, y, z) \in \mathbb{R}^{128 \times 128 \times 128}$$

In order to have each modality standardized Z-scoring is used to independently standardize each one. As seen below.

$$\mu_m = E[\tilde{I}_m], \sigma_m = \sqrt{V[\tilde{I}_m]} \quad (7)$$

$$\hat{I}_m = \frac{\tilde{I}_m - \mu_m}{\sigma_m} \quad (8)$$

Once the modalities are standardized, they are channel stacked to create a 40channel 3D tensor used as an input to the segmentation network.

$$X = [\hat{I}_1 \ \hat{I}_2 \ \hat{I}_3 \ \hat{I}_4] \in \mathbb{R}^{4 \times 128 \times 128 \times 128}$$

This preprocessing ensures consistent resolution and intensity scaling necessary for stable CNN training.

3D Tumor Segmentation Network

A 3D U-Net-based architecture is used to segment the MRI volume into four anatomical regions. These regions are the background, necrotic/non-enhancing core (NCR), edema (ED), and enhancing tumor (ET). The model that was developed consists of an encoder decoder structure with skip connections enabling recovery of spatial detail lost during down sampling. The following shows how this architecture was setup.

Given an input tensor X , the network predicts a voxel-wise segmentation mask

$$\hat{Y}(x, y, z) \in \{0, 1, 2, 3\} \quad (9)$$

An Attention based U-Net was selected because skip connection gating improves detection of small, low-contrast tumor subregions an important engineering optimization over standard U-Net models.

Each encoder block contains two 3D convolutions followed by ReLU activations and a 3D max-pool.

$$E_{k+1} = \text{Pool}(\text{ReLU}(W_{k,2} * \text{ReLU}(W_{k,1} * E_k))) \quad (10)$$

From here each decoder blocks up samples the feature maps and connects them with the corresponding encoder features.

$$D_k = \text{Conv}(\text{Up}(D_{k+1}) \parallel E_k) \quad (11)$$

“ $\parallel$ ” represents channel connecting.

For the final prediction layer the segmentation logits are produced using a $1 \times 1 \times 1$ convolution.

$$\hat{Y} = \text{argmax}(\text{Conv}_{1 \times 1 \times 1}(D_1)) \quad (12)$$

Radiomic Feature Extraction

After segmentation has been completed, all the meaningful radiomic features are derived from the tumor mask. Following segmentation, biologically meaningful radiomic features are derived directly from the tumor mask. [1] The BraTS labels are 1 necrotic/non-enhancing tumor (NCR/NET), 2 peritumoral edema (ED), and 4 enhancing tumor (ET). The predicted mask is represented by $M(x, y, z)$. The volumetric features are. Where Whole Tumor (WT), Tumor Core (TC), Enhancing Tumor (ET), and Edema (ED) form the radiomic signature used for survival modeling.

$$\text{WT} = \sum_{x,y,z} \mathbf{1}[M > 0] \quad (13)$$

$$\text{TC} = \sum_{x,y,z} \mathbf{1}[M = 1] + \mathbf{1}[M = \text{ET}] \quad (14)$$

$$\text{ET} = \sum_{x,y,z} \mathbf{1}[M = \text{ET}] \quad (15)$$

$$\text{ED} = \sum_{x,y,z} \mathbf{1}[M = 2] \quad (16)$$

These features correspond directly to physical tumors, which is clinically meaningful for surgical planning and prognosis.

Survival Regression Network

Finally for the survival regression it's a fully connected neural network to predict the survival time from the radiomic and clinical data. The input feature vector for each patient is as shown.

$$f = [\text{age}, \text{resection}, \text{WT}, \text{TC}, \text{ET}, \text{ED}] \quad (17)$$

The final prediction in days is obtained by rescaling, where $S_{\max}$ is the maximum survival in the training dataset.

$$\hat{s} = \hat{y} \cdot S_{\max} \quad (18)$$

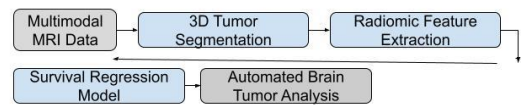


Figure 0 Brain Tumor MRI Analysis Pipeline

The survival network contains 2,561 trainable parameters, computed as.

fc1.weight $\rightarrow$ 384

fc1.bias $\rightarrow$ 64

fc2.weight $\rightarrow$ 2048

fc2.bias $\rightarrow$ 32

fc3.weight $\rightarrow$ 32

fc3.bias $\rightarrow$ 1

Results

The deep learning pipelines consisted of two major components, those being a 3D Attention U-Net for brain tumor segmentation and a fully connected survival prediction network that used radiomic features extracted from the models output along with additional patient data.

The Attention UNet was trained for 25 epochs using the Dual-Focal loss function. Figure 1 illustrates the evolution of the training and validation loss across epochs, while Figure 2 shows the corresponding validation Dice score. As expected, the training loss decreases smoothly from 0.40 to about 0.09, indicating stable optimization. Validation loss shows mild oscillations, which can be common in medical segmentation tasks, where there is some class imbalance. Despite these fluctuations, the validation Dice score remained consistent within the 0.55 – 0.65 range, demonstrating that the model was able to learn meaningful tumor structures throughout training.

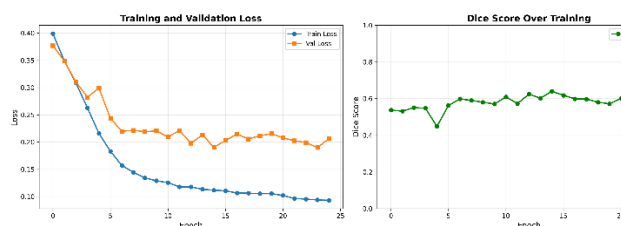


Figure 1 and 2 Training and Validation Loss and Dice Score

The model performed the best when it came to identifying the Whole Tumor, with a Dice Score of 0.79 as seen in figure 10. When it came to the Edema it had reliable segmentation of the surround swelling region with a dice score of

0.60. The enhancing tumor and the tumor core had dice values around 0.52-0.55, which is consistent with the small and more heterogeneous nature of these subregions. On the other hand, the necrotic area was the most challenging region due to the irregular shapes and low contrast with a dice score of 0.42. These metrics collectively confirm that the model produces clinically meaningful anatomical segmentations across all major tumor substructures. Figures 3-5 present the qualitative segmentation samples for epochs 15 through 25, including all four MRI modalities, the ground truth mask, and the model's prediction. Across the epoch, several trends can be observed. For example, there was an improved delineation of tumor borders, especially in the edema and enhancing regions. After epoch there was sharper boundary definitions and a reduction in false positives. By epoch 25, the predicted segmentation visibility aligns with the ground truth, capturing the major tumor compartments with strong consistency.

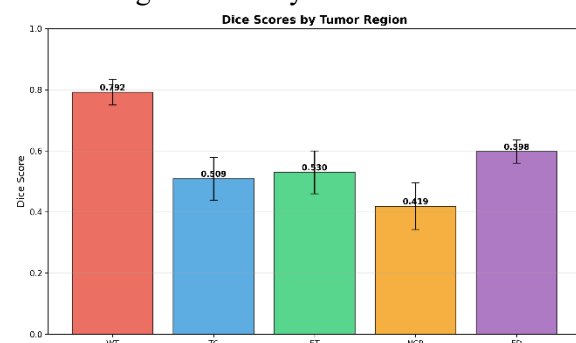


Figure 10 Dice Scores by Tumor Region

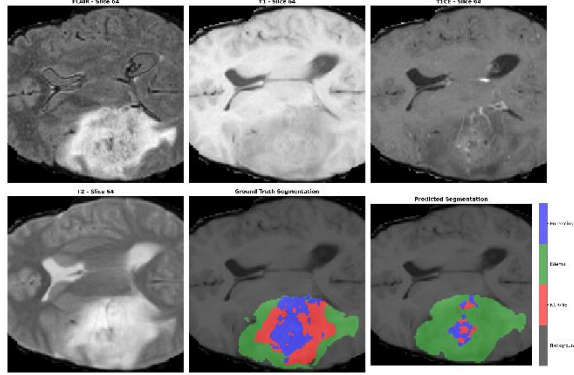


Figure 3 Epoch 15 Prediction Segmentation

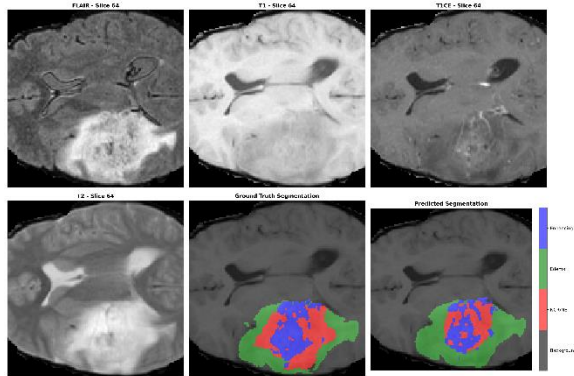


Figure 4 Epoch 20 Prediction Segmentation

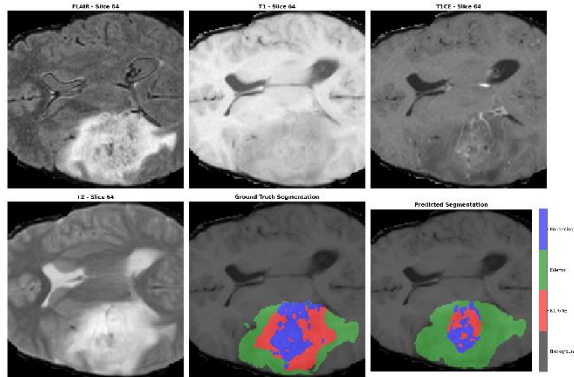


Figure 5 Epoch 25 Prediction Segmentation

After extracting radiomic features from the segmentation masks, which include the whole tumor volume, tumor core, enhancing tumor, edema, patient age, and resection status, the fully connected survival prediction network was trained to get an estimate of the overall survival in days. The model was trained on 234 patients who had complete survival information. Figure 7

shows the Kaplan Mier survival curves stratified by surgical resection status. Where GTR is the Gross Total Resection, STR the Subtotal Resection and the Unknown. Unknown being where the resection wasn't fully clarified in the data. These curves followed expected clinical patterns. Patients receiving GTR generally exhibit longer survival, whereas STR patients showed earlier decline in survival probability. The unknown group spans a mixture of surgical outcomes but falls between the two curves. This confirms that the dataset reflects real clinical difference in survival and provides a structure for the prediction model.

Figure 6 illustrates the model predicted survival days vs true survival data for all the patients in the validation set. Where the red dashed line represents a perfect 1:1 correspondence. Most predictions clustered near the trend line, indicating that the network captures the general survival distribution effectively. The model tends to slightly underpredict very long survival cases >1000 days, this is a known limitation in neural regressors when the distribution is heavily skewed. Nevertheless, the correlation remains strong across the clinically relevant 0-800 day range.

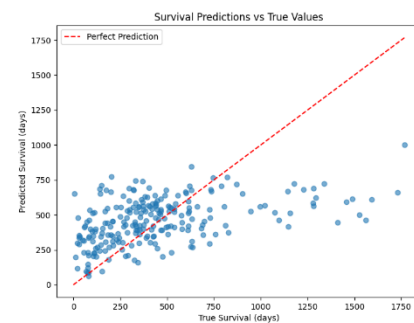


Figure 6 Survival Predictions Vs True Values

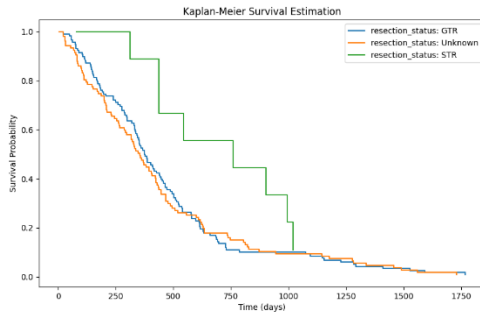


Figure 7 Kaplan Meier Survival Estimation

Once trained the full pipeline was applied and tested as can be seen in Figure 8 where with the age of 54.3 a survival prediction was expected of 500 days, however their actual survival was 464 days.

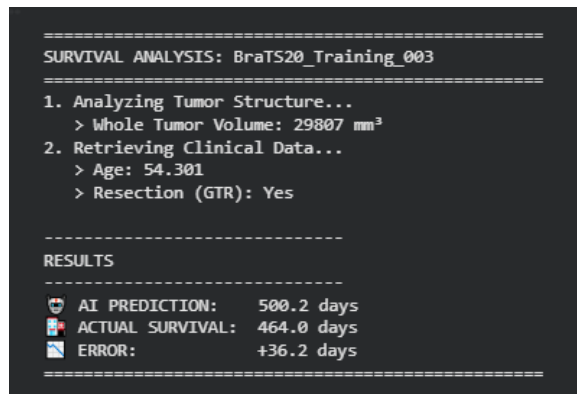


Figure 8 Survival Prediction

Conclusion

In this work, a complete engineering pipeline was developed for automated analysis of glioma MRI scans integrating 3D tumor segmentation with radiomic survival prediction. Using multimodal MRI inputs and a 3D Attention UNet architecture the system was able to generate meaningful segmentation masks across all major tumor subregions, including the whole tumor, edema, enhancing tumor, and the necrotic core. The resulting radiomic features combined with patient age and resection status were used to train a fully connected regression model to estimate patient survival time.

The segmentation network demonstrated stable coverage and produced clinically relevant Dice scores with the strongest performance observed for the whole tumor and edema regions. The survival prediction model captured the overall relationships between tumor morphology and patient outcome, as reflected in the Kaplan Meier survival trends and the alignment between predicted and true survival times. These results collectively showed the developed system effectively translates raw medical imaging signals into quantitative bio marks and predicted indicators, which highlight its value as an engineering tool for medical decision support.

Several engineering challenges were encountered throughout the project. Segmentation accuracy for small or low contrast tumor subregions remains limited due to some class imbalance and heterogeneous imaging conditions. Survival prediction performance was also constrained by some missing clinical variables, along with that the skew distribution of long term survival. Computational constraints also restricted the training of deeper or more data intensive models. These limitations show clear directions for improvement in this model.

Despite these challenges, a successful pipeline was established to create a strong foundation for future engineering development in automated neuro-oncology analysis. Future work could incorporate more imaging, or other architectures for improved generalization. Clinically, the system could support pre-operative planning, risk stratification, and treatment decision workflows by providing rapid assessment of tumor burden and expected survival. Thus, this work demonstrates the potential of integrated machine learning pipelines to help enhance medical imaging analysis and deliver meaningful engineering contributions to modern healthcare applications.

References

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Code

Refer to comments on how to run model

https://colab.research.google.com/drive/1YmUXJmJSkP4OdnS1_OxQ_oRZt9As7xRq?usp=sharing